

EDUCATION

University of California San Diego (UCSD)

B.S. Mechanical Engineering (MAE), Minor in Cognitive Science Design

2023 - June 2027 (expected)

Cumulative GPA: 3.37

WORK EXPERIENCE

Rocket Lab USA

September 2026 - Present

Mechanical Engineering Intern - Harness Design

- Engineered RTD sensor **harness** wiring diagram in **Capital Essentials**, defined 50+ inline connections in space vehicle master schematic. Executed **CAD electrical routing** in **Siemens Nx** and generated **formboard technical drawings** to direct physical manufacturing. Collaborating towards routing final CAD assembly for heater and RTD sensors within harness kit.
- Automating harness CAD workflows by using **Python** scripting, traversing wiring diagrams to streamline CAD assembly process with mapped ports and ID instances to optimize constraints and formboard drafting (cut list, BOM, resistance measurements, etc.)
- Establishing standardized procedure and building space vehicle mockups to conduct fit-check testing and harness integration.

Aynak, Inc.

January 2025 - August 2026

Mechanical Engineering Intern

- Spearheaded design and hardware integration of 0.11 mm thin speaker **flex PCBs**, optimizing stack-up thickness and routing through 3 mm heat-set hinges within smart-eyewear frames housing 4 development boards and 1 battery.
- Designed and **SLA** printed custom mounts for 3 distinct speaker configurations, used in experimental testing of beamformer sound transmission. Optimized mount geometries and hardware placement to minimize howling and acoustic feedback.
- Prototyped hearing aid receiver mount integrated into smart eyewear frames of sub-3 mm internal clearance.
- Performed quality assurance and testing of actuators for bone conduction hearing within rigid components of eyewear frame.

Luan Lab, University of California San Diego

January 2025 - June 2026

Undergraduate Researcher <https://sites.google.com/ucsd.edu/luangroup>

- Implementing mechanical guided assembly into wearable and flexible electronics such as skin sensors and biofeedback mechanisms. Heading experimental procedures on morphing systems and programmable structures.
- Utilized Gallium-Indium synthesis approach for electromagnetically reconfigured channels within substrates.
- Fabricated 5 mm wide substrates via **spincoating** and 4 custom **PLA** mold patterns designed in **SolidWorks**.
- Aiming to develop novel electromagnetism-based methods for patient specific morphing healthcare devices.

MAE 3 Course Tutor

March 2025 - June 2026

Design Studio Tutor

- Redesigned 1 of 2 project fabrication procedures for **Fusion 360 CAD** of pendulum clock, developing smoothed replacement **CAD** models, creating labeled GIF graphics and video tutorials, and modeling common errors with prevention strategies.
- Constructed "World cup" robot combat arena in AutoCAD, lasercut and assembled with T-Slot frames and material constraints.
- Designed a geared motor adapter and demo for ongoing lab curriculum to demonstrate press fit and linkages in lecture.
- Maintained fabrication studio, assist in instrumentation troubleshooting, and maintain **LaserCMM** and **Bambu 3D printers**.

Instructional Assistant, Lab Tutor

- Provided mentorship to a total of 19 students on **Fusion 360**, **AutoCAD**, and **GD&T**, leading 2 fabrication projects in studio.
- Taught manufacturing, fabrication, and **CAD** skills, including **Fusion 360**, **AutoCAD**, laser cutting, and **3D printing**.

Vi at La Jolla Village Senior Living

July 2023 - April 2026

Server. Adapted to dual roles as server and hostess

RELEVANT PROJECTS

Triton Robotics

January 2024 - June 2025

UCSD Robotics Team, Infantry Mechanical Division | tritonrobotics.org

- Supported Triton Robotics' first time hosting RMNA competition, manufacturing playing fields and obstacles for 28 universities.
- Redesigned armor mount with interference fit for **ease of assembly** and time efficiency during competition. Integrated a **3D printed (PLA)** snap-fit joint to mount and accommodate new slip ring using **OnShape**.
- Developed a **SolidWorks** model Omni wheel into a new suspension system, optimizing it for a new motor assembly of chassis.
- Initiation project: designed bearing armor cover with rotational functionality to serve as protection during competition

Portable Volleyball Flywheel

Summer 2025

SKILLS

Technical Skills: GD&T, DFM, machine shop, Python (Pandas, Tkinter), SLA/FDM 3D printing, flex PCB integration

Software & Tools: Siemens Nx CAD (Electrical routing), Teamcenter, Capital Essentials, SolidWorks (CAM, motion analysis, FEA), Fusion 360 (Mesh, Surface, Sheet Metal), AutoCAD, ANSYS (FEA), MATLAB, Excel, Visual Studio Code